

Note: ChatGPT was used to create these responses. The chatbot was told to take on the following persona: A student in the course who is interested in using generative AI to enhance personalized learning in experimental methods for aerospace engineering. The chatbot was instructed to pause after each stage. This output is provided for learners as an example of the process and/or in the event they do not have access to a chatbot.

1. Define Your Goal or Problem

My goal: I want to explore how generative AI can support personalized learning within an aerospace engineering laboratory course, specifically the Experimental Methods course that many students struggle with. As a graduate TA and as someone who wants to eventually teach this material, I've noticed that students often repeat the same conceptual mistakes—uncertainty propagation, calibration curves, dynamic response, and instrumentation selection—but they make these mistakes for different reasons. A single explanation rarely works for everyone.

Why AI fits this task: Generative AI seems well suited to this problem because it can provide:

customized explanations,

tailored examples,

adaptive difficulty,

and instant feedback based on the learner's unique misunderstanding.

In other words, AI can help create a personalized tutor that adjusts explanations depending on whether a student is visual, hands-on, conceptual, or math-oriented. I also think AI can help instructors generate these personalized materials efficiently.

Prompt to send: “Hi! I’m working on a project where I want to use generative AI to support personalized learning in an aerospace engineering experimental methods course. I’m imagining something like an AI tutor that adapts explanations to different student learning styles (visual, conceptual, step-by-step, or big-picture). Can you help me refine this idea into a clearer, more focused goal that I could realistically explore in a short mini-capstone project?”

2. Choose and Use Two AI Tools

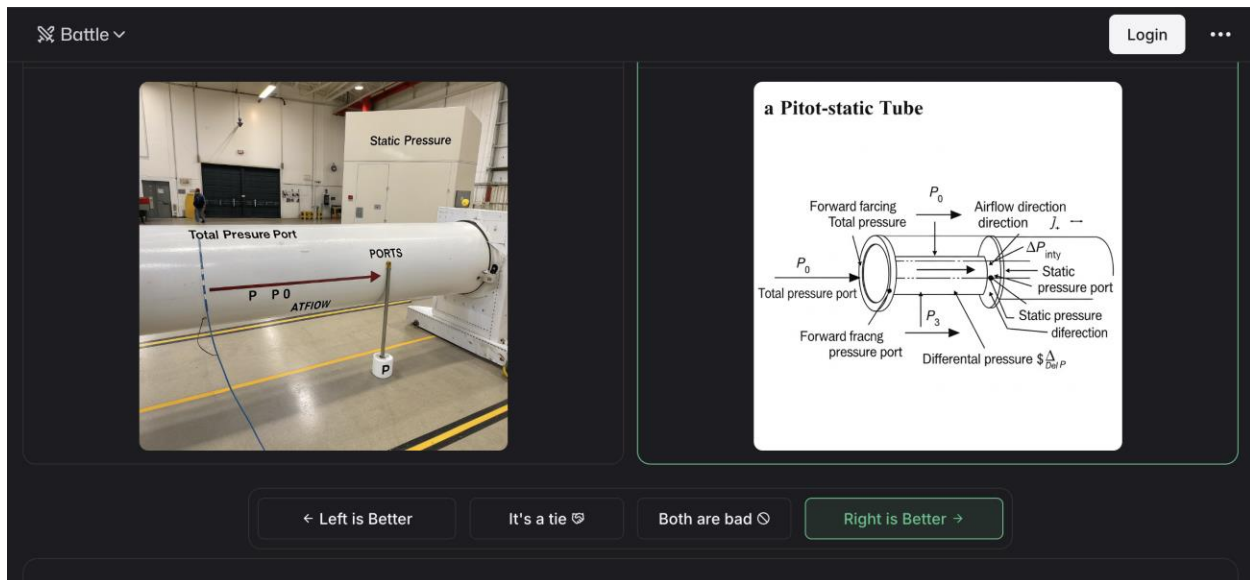
For my second tool, I used a **multimodal generative AI platform** (I tested it through LM Arena, which aggregates several image models). My goal was to generate a simple, instructional-style diagram of a Pitot-static tube—something resembling what we use in our aerospace lab manuals.

The tool successfully generated clean, high-resolution images, each showing variations of a Pitot-static tube with airflow arrows and pressure labels. While the images were visually appealing, they weren't accurate enough for real instructional use. For instance, some models added unrealistic port placements or stylized shapes that wouldn't appear on an actual Pitot tube. Still, they communicated the general idea well enough for this exercise and demonstrated how multimodal AI tools can supplement text-based explanations.

Representative Output (summary):

- A 3D-rendered tube pointing into oncoming arrows labeled “airflow.”
- A forward-facing central port labeled “Total Pressure” or P_{0P_0} .
- Several side holes labeled “Static Pressure” or $P_{\infty P_{\infty}}$.
- A ΔP label showing the difference between the two pressures.
- Overall aesthetic similar to a technical sketch, but not precise enough for real lab instruction.

This helped me see how multimodal AI can quickly create visuals to support personalized explanations—but also revealed the limitations when precision or domain accuracy matters.



3. Compare and Reflect on the Process

Using two different AI tools gave me a much clearer understanding of both the strengths and limitations of generative AI in a technical education context. The text-based chatbot was surprisingly effective at producing helpful, customized explanations. The four learning-style variations felt genuinely different, and the chatbot generated new angles, metaphors, and structures I wouldn't have thought of on my own. The process of getting useful ideas was actually easier and more productive than I expected. With only a single prompt, I received a concise set of personalized explanations that could realistically support different types of students.

The multimodal image generation tool was a very different experience. From an artistic or aesthetic standpoint, the images were *excellent*: clean, attractive, high-resolution, and visually engaging. However, they were not usable in a technical aerospace setting. Even though the tool captured the *concept* of a Pitot-static tube—forward port, side ports, airflow arrows—the specific geometry, port placement, and proportions were wrong. These inaccuracies mean the images could confuse students or misrepresent how the hardware actually works. In some cases, the tube resembled a stylized sci-fi probe more than real instrumentation.

One surprise was that the multimodal tool excelled at the parts that *didn't need to be accurate* (composition, style, color, clarity) but struggled with the parts that *absolutely must be accurate* (hardware details, engineering geometry). The chatbot, on the other hand, had the opposite behavior—it delivered surprisingly high-quality conceptual

reasoning but doesn't produce visual fidelity. This contrast made me appreciate that generative AI can be powerful, but only when matched to the right task.

Overall, the text tool was easier to control, while the image tool required more trial and error and still never produced technically correct results. The experience reinforced that AI tools can support idea generation and early-stage communication but still need careful human oversight for accurate technical content.

4. Identify Ethical and Practical Considerations

Working with these tools made it clear that using generative AI in an engineering education context raises both ethical and practical concerns. The biggest issue I encountered was the risk of **misinformation through confident but incorrect visuals**. The image models produced diagrams that looked polished and “official,” but contained technical inaccuracies that could mislead students learning about Pitot-static instrumentation for the first time. This is especially problematic in aerospace, where misunderstandings about measurement systems can cascade into incorrect analysis or safety-critical errors. An AI-generated image that *looks right but isn't* could easily be taken at face value by a novice learner.

On the text side, the explanations were solid, but I still noticed the potential for **overgeneralization and subtle conceptual shortcuts**. The chatbot's descriptions were plausible and helpful, but without careful review, a learner might assume every detail is precise. For instance, the analytical explanation referenced typical alignment tolerances and Mach constraints; although accurate, these values might not apply to every lab environment. This highlights the need for domain oversight: AI can scaffold understanding, but it shouldn't act as an authoritative source without verification.

Another ethical concern is **responsibility and transparency**. If I were to use an AI-driven personalized tutor in a real class, I would need to be clear with students that the explanations come from a probabilistic model, not an instructor-reviewed source. Students deserve to know when content is AI-generated, particularly when it might contain inaccuracies. There is also a privacy dimension—some personalized learning systems require uploading course documents or student responses. If not handled carefully, that raises questions about data ownership, model training, and FERPA-style considerations.

To address these issues, I would:

- Use AI text generation only as a *drafting tool*, with instructor or TA review before giving it to students.
- Avoid using AI-generated diagrams for technical instruction unless they're thoroughly corrected.
- Make it explicit to learners what parts of the content are AI-assisted.
- Keep all uploaded or retrieved materials institutionally controlled and free from sensitive or copyrighted lab manuals unless permission is secured.

In short, generative AI is a powerful support tool, but it still needs human interpretation, ethical awareness, and quality control—especially in an educational setting where accuracy matters.

5. Evaluate Outcomes

Overall, generative AI absolutely helped me make progress toward my goal, but not in the exact way I initially expected. The text-based chatbot was genuinely useful for generating personalized learning materials. It produced explanations that were clear, well-structured, and noticeably different across learning styles. If I were building an AI tutor prototype, this would give me a strong foundation to work from. It saved me time, expanded the range of explanation styles I could offer, and helped me think about the content from perspectives I probably wouldn't have considered on my own.

However, the AI did struggle in some important ways. The image generation tool produced visuals that were aesthetically polished but technically wrong. This reinforced a major limitation: generative AI models can “look right” while being fundamentally inaccurate. In an engineering course—where diagrams of instrumentation must be physically correct—this is a serious drawback. Even the text model sometimes glossed over important caveats (like compressibility effects or port sensitivity) that a human instructor would naturally clarify.

Another limitation was the AI's lack of self-awareness about its own uncertainty. The chatbot delivered explanations confidently, even when it introduced assumptions that wouldn't hold for all wind-tunnel setups. This means that while the AI was a great

brainstorming partner, I had to stay in the loop as the technical verifier. The tools weren't ready to independently generate instruction-ready content without review.

Still, I would absolutely use these tools again—just not as final authorities. I would trust a text-based model for:

- generating first-draft explanations,
- offering multiple learning-style versions quickly,
- brainstorming alternative metaphors or visualizations, and
- helping students gain intuition in low-stakes contexts.

But I would *not* trust these tools for:

- producing technical diagrams,
- providing unreviewed numerical or procedural guidance, or
- conveying safety-critical information.

The biggest takeaway from my evaluation is this:

AI accelerates the creative and conceptual parts of personalized learning, but human oversight is still essential for accuracy.

Used intentionally, AI can make teaching and learning more adaptive and more efficient—but it has to be paired with engineering judgment.

6. Summarize What You Learned

This project helped me see generative AI less as a replacement for expertise and more as a partner in the learning process. I learned that AI is extremely good at generating diverse explanations, adapting tone and structure to different types of learners, and giving me creative angles I hadn't considered. In that sense, it truly supports personalized learning by widening the range of teaching approaches available to an instructor or tutor.

At the same time, the project highlighted clear limitations. The tools can sound confident even when they simplify or slightly distort technical details, and the multimodal image models, while visually impressive, produced diagrams that were unusable in an actual aerospace setting. This reminded me that AI can accelerate understanding, but it cannot guarantee accuracy. Human judgment, verification, and domain knowledge remain essential—especially in engineering, where details matter.

I also learned something about myself as an AI user: I tend to trust well-written explanations, and it took conscious effort to slow down and critically evaluate what the model produced. This project reinforced that my role is not just to consume AI output, but to shape it, refine it, and check it. The value humans bring—context, caution, technical insight, and ethical responsibility—is what turns AI-generated content into something meaningful and reliable.

My overall takeaway is that AI can be a powerful collaborator if used thoughtfully. It helps with idea generation, explanation diversity, and early-stage learning support, but it still needs a human partner to guide, filter, and validate the results. That partnership—combining AI's adaptability with human judgment—is what makes personalized learning in engineering both possible and responsible.